

ENGR 1451/2451: Exploratory Data Science

Normal Approximation

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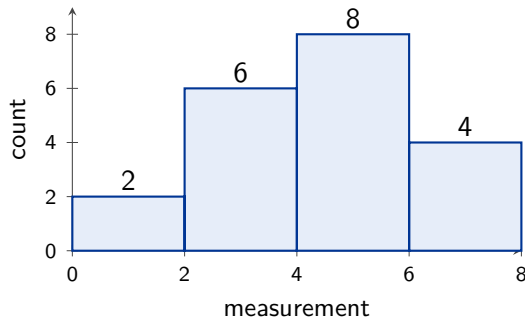
Today's Goals

Use the observed sample to build and assess a normal approximation.

- Normalize a histogram with **equal-width bins**.
- Fit a normal curve using the sample mean $\bar{x}$ and sample standard deviation s .
- Use Python to estimate percentages and percentiles.
- Compare the fitted curve with actual sample proportions.
- Distinguish raw observations from repeated sums and averages.

Start with Equal-Width Bins

- One small teaching sample: $n = 20$ measurements.
- Bin edges: 0, 2, 4, 6, 8.
- Every bin has the **same width**, $w = 2$.
- Bar height is the **number of observations** in that bin.



Count scale

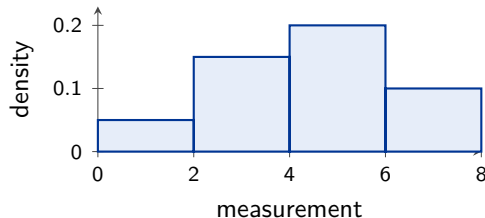
Counts add to the sample size, not to 1.

Normalize Counts to Density

For a common bin width w , normalize in two steps:

$$\underbrace{n_i}_{\text{count}} \xrightarrow{\div n} \underbrace{p_i = \frac{n_i}{n}}_{\text{proportion}} \xrightarrow{\div w} \underbrace{h_i = \frac{n_i}{nw}}_{\text{density}}.$$

Bin	Count	Proportion	Density
$[0, 2)$	2	0.10	0.05
$[2, 4)$	6	0.30	0.15
$[4, 6)$	8	0.40	0.20
$[6, 8]$	4	0.20	0.10

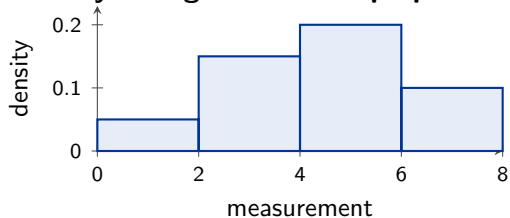


Normalize the area, not the heights

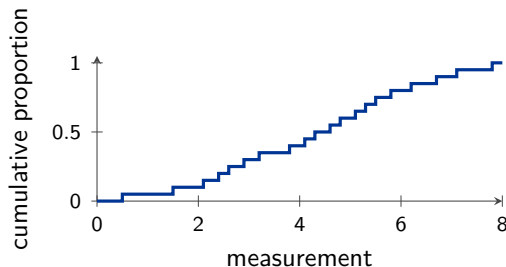
bar area = $h_i w = p_i$, so total area is 1.

Density and Cumulative Sample Proportion

Density histogram: area is proportion



Empirical CDF: count up to a value



Two ways to describe the observed sample

A histogram groups observations into bins. The empirical CDF counts each observation:

$$\hat{F}_{\text{sample}}(x) = \frac{\#\{x_i \leq x\}}{n}.$$

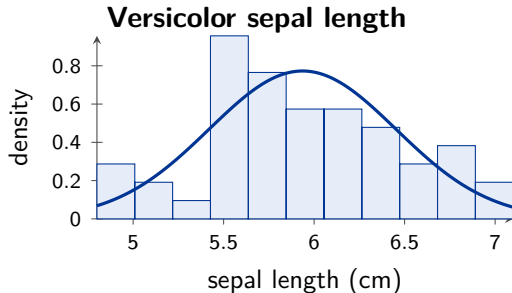
Fit a Normal Curve to the Sample

Observed sample: measurements, histogram, and empirical proportions.

Fitted normal curve: a smooth approximation centered at $\bar{x}$ with spread s .

$$\hat{f}(x) = \frac{1}{s\sqrt{2\pi}} \exp\left[-\frac{(x - \bar{x})^2}{2s^2}\right].$$

Use $s > 0$. This is the curve's definition; Python evaluates it.

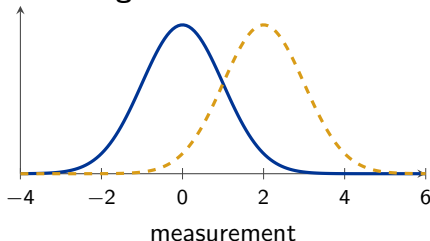


A fit is not a fact about the data

A normal curve may not reproduce skewness, subgroups, or unusual observations.

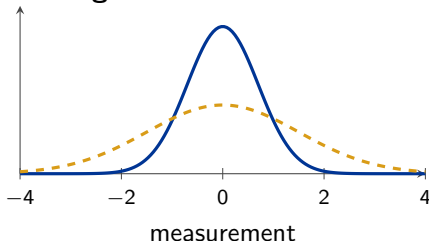
The Sample Center and Spread Set the Curve

Change $\bar{x}$: shift the curve



Same s : same shape, different center.

Change s : widen or narrow it



Larger s : wider and lower peak.

Every fitted density curve has total area 1

Changing center or spread does not change the total area.

Standard Units and Changes of Scale

Standardize relative to the sample

$$z_i = \frac{x_i - \bar{x}}{s}.$$

- Positive: above the sample mean.
- Negative: below the sample mean.
- Zero: at the sample mean.

Change measurement units

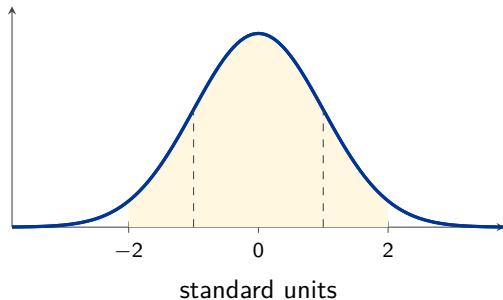
For $y_i = ax_i + c$ with $a > 0$:

$$\bar{y} = a\bar{x} + c, \quad s_y = as_x.$$

Standardized positions are unchanged.

Example in Python: convert Iris measurements from centimeters to millimeters.

Normal-Curve Areas: Useful Benchmarks



Approximate fitted areas

68% within $\bar{x} \pm s$

95% within $\bar{x} \pm 2s$

99.7% within $\bar{x} \pm 3s$

The actual sample percentages can differ.

Python Calculates Areas and Cutoffs

Create a fitted normal curve using `norm(loc=mean_value, scale=std_value)`.

Question	Meaning	Python method
Density at x	Fitted density height	<code>model.pdf(x)</code>
At or below x	Area to the left of x	<code>model.cdf(x)</code>
Above x	Area to the right of x	<code>model.sf(x)</code>
Between a and b	Area between two endpoints	<code>model.cdf(b) - model.cdf(a)</code>
Cutoff with p to the left	Value at percentile p	<code>model.ppf(p)</code>
Cutoff with p to the right	Value with right-tail area p	<code>model.isf(p)</code>

Remember the pairs

`cdf`: $x \rightarrow$ left area

`ppf`: left area $\rightarrow x$

`sf`: $x \rightarrow$ right area

`isf`: right area $\rightarrow x$

`sf` = **survival function**; `isf` = **inverse survival function**; `ppf` = **percent point function** (inverse CDF).

Normal Approximation: Inspect, Fit, Compare

1. Inspect the sample

Plot equal-width bins.
Check shape, unusual observations, and subgroups.

2. Fit a normal curve

Calculate $\bar{x}$ and s from the sample. Use them as the curve's center and spread.

3. Compare answers

Use Python for fitted areas and percentiles. Compare with counts and quantiles from the sample.

Two different answers are useful

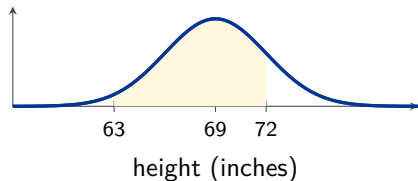
Observed proportion: count the sample values in the interval.

Fitted estimate: use the area under the normal curve.

Height Examples: Let Python Find the Areas

Men: between 63 and 72 inches

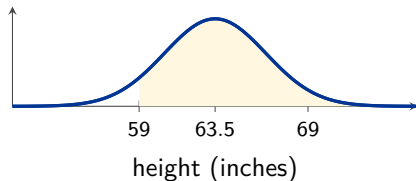
$$\bar{x} = 69 \text{ in.}, \quad s = 3 \text{ in.}$$



Python fitted estimate: **81.86%**

Women: above 59 inches

$$\bar{x} = 63.5 \text{ in.}, \quad s = 3 \text{ in.}$$



Python fitted estimate: **93.32%**

Percentiles: Data Cutoff or Fitted Cutoff?

Percentile from the sample

Use the ordered observations.

```
sample.quantile(0.95)
```

The result depends on the observed values and interpolation convention.

Percentile from the fitted curve

Find the cutoff with 95% of fitted area to its left.

```
model.ppf(0.95)
```

The result depends on the fitted center and spread.

SAT example from the earlier slides

Use the illustrative summaries $\bar{x} = 535$ and $s = 100$. Python returns a fitted 95th-percentile cutoff of **699.5**, or about 700.

Chapter 6 Applications, Revisited in Python

Reuse the numerical settings as **illustrative sample summaries**.

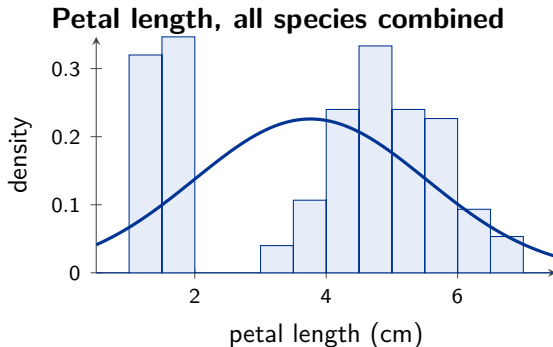
Example inputs	Question	Method
Current: $\bar{x} = 10$ mA, $s = 2$ mA	Percentage above 13 mA	
Assembly: $\bar{x} = 20$ h, $s = 2$ h	Percentage below 19.5 h; between 20 and 22 h	
Cutoffs: $\bar{x} = 20$, $s = 9$	Value with 19% to the left or 32% to the right	

Check the Sample Shape First

- A normal curve is most useful for one roughly symmetric mound.
- Check for long tails, outliers, and distinct groups.
- Compare fitted percentages with the observed sample.
- A larger sample does not repair a poor choice of shape.

Iris example

Combining species hides structure that one normal curve does not capture well.



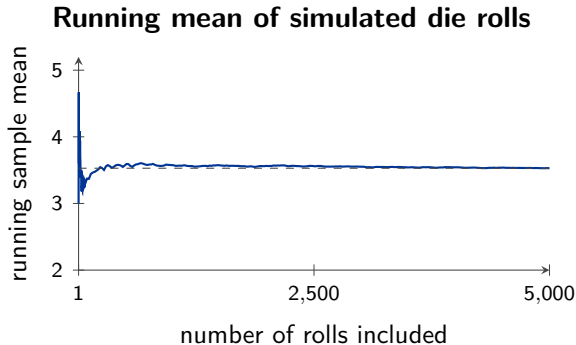
Companion notebook: Section 10: compare the combined histogram with separate species.

Law of Large Numbers: Averages Stabilize

For independent observations from the same process with a finite mean, the running average settles as more observations are included.

This is about stability, not normality.

More individual die rolls still give a roughly flat histogram of 1 through 6.



Central Limit Theorem: Repeated Sums and Averages

Repeat a group of m independent observations from the same process, then record:

$$S_m = X_1 + \cdots + X_m, \quad \bar{X}_m = \frac{S_m}{m}.$$

What becomes approximately normal?

The distribution of **repeated sums or averages**, for sufficiently large m and finite, nonzero variance.

A group of raw observations need not itself look normal.

Use sample information for the spread

If s describes individual observations:

$$\text{estimated spread of means} \approx \frac{s}{\sqrt{m}},$$

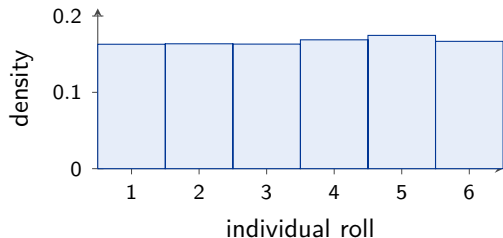
$$\text{estimated spread of sums} \approx s\sqrt{m}.$$

These are estimates, not exact identities for one sample.

There is no universal group size that guarantees a good normal approximation.

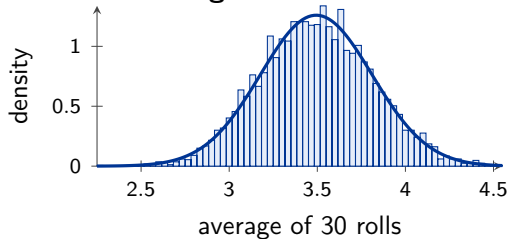
CLT in Python: Rolls versus Group Averages

One roll per experiment



Roughly flat across the six outcomes.

Average of 30 rolls



Approximately bell-shaped and less spread out.

The sum and the average tell the same shape story

The sum has a larger center. The average remains near the individual-roll sample center. Rescaling relates the two.

Why Can Individual Measurements Look Normal?

Many effects within one observation

A conceptual measurement model is

$$\text{measurement} = \text{baseline} + E_1 + \cdots + E_k.$$

Many small, roughly independent additive effects can produce a bell-shaped sample.

Finite variances and no dominating effect make this explanation plausible, not automatic.

Many observations in a dataset

Collecting more observations reveals the sample's shape more clearly.

It does not make a uniform or skewed pattern become normal.

To see the CLT, repeatedly form **sums or averages** of groups.

What matters is how a measurement is generated, not just how many measurements are collected.

Normal Approximation